Lesson 15: Ready to Show

Unreal Engine (optional advanced track)

VIDEO GAME CREATION

Today has no new code to build. It's a workshop day: polish what's roughest, then present your finished capstone game to the class. This is the Level 3 Refined milestone - the end of Semester 1's build.

Open your own [YourName]Game project from Lesson 14 - this IS your capstone. There's nothing separate to submit.

Your Full Capstone Checklist

Work through this checklist first. Anything unchecked is your polish-time priority - fix these before adding anything new.

Core Mechanics (Lessons 2-11)

- ☐ Move and shoot respond correctly to input, every time.
- ☐ Enemies are destroyed on hit, and score increases correctly.
- ☐ An enemy escaping unshot costs a life, and lives are tracked correctly.
- ☐ Score, lives, and wave are all shown clearly on screen at all times.
- ☐ Winning and losing both work, and both show a clear message with a restart hint.
- ☐ Pressing R restarts cleanly from any state - mid-round, after a win, or after a loss - back to wave 1.
- ☐ Multiple waves work correctly: clearing a non-final wave spawns a fresh one; only the final wave triggers a real win.

Sound (Lesson 12)

- ☐ A shot makes a sound.
- ☐ A hit/explosion makes a sound.
- ☐ A wave clear makes a sound, distinct from the win sound.
- ☐ A game over makes a sound.

Animation (Lesson 13)

- ☐ A hit briefly flashes the enemy before it's removed.
- ☐ An independent explosion effect appears and cleans itself up on every hit.

Difficulty (Lesson 14)

- ☐ Enemy count increases with each wave.

- ☐ Enemy speed increases with each wave.
- ☐ Your chosen Level 3 stretch feature is present in at least a rough, good-faith form.

Polish Pass

With whatever time remains after the checklist above, work top to bottom through these polish priorities - in order:

1. Fix anything broken or unchecked above first - a complete, working Level 1/2 game beats a flashy but broken one.
2. Tune anything that feels unfair or confusing - difficulty too easy/hard, unclear on-screen text, a sound that's too loud or jarring.
3. Only after 1 and 2: continue building out your Lesson 14 stretch feature, or add a small extra touch if you have time and energy left.

- Common Unreal rough edges to check: a Custom Event dispatcher bound twice (causing double-counted hits), an unset Blueprint variable defaulting to zero/null, or a Delay node left from testing that makes the game feel sluggish.

Preparing Your Presentation

Presentations are short - about 2 minutes each. You don't need slides or notes; just be ready to talk while you play. A simple structure that works well:

4. Name your game and describe its core mechanic in one sentence.
5. Play a short round - enough to show a hit, a wave clear, and either a win or a loss.
6. Point out your chosen Level 3 stretch feature specifically, even if it's not fully finished.
7. If something breaks live, that's genuinely fine - briefly say what it's supposed to do and move on. A confident presentation matters more than a perfect one.

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